

Linzhen Zhu

Ph.D. Candidate | University of Michigan, Ann Arbor

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EDUCATION

University of Michigan, Ann Arbor

Sep. 2025–Present

Ph.D. Candidate in Computer Science and Engineering

Advisor: Prof. Ke Sun, Ambient Intelligence (AmI) Lab

University of Michigan, Ann Arbor

Apr. 2026

Master of Science in Engineering (Computer Science and Engineering)

University of Nottingham Ningbo China

Jul. 2024

B.Eng. in Electrical and Electronic Engineering, First Class

RESEARCH INTERESTS

Optical and tactile sensing; mobile and ubiquitous computing; sensor–actuator systems; embedded and wearable hardware; human–computer interaction.

RESEARCH EXPERIENCE

Optical and Tactile Sensing — University of Michigan

Advisor: Prof. Ke Sun, Ambient Intelligence Lab

- **MoiréSkin:** First-author work on Moiré-based tactile sensing with pneumatic actuation and adjustable compliance. *ACM MobiCom 2026*, full paper and demo accepted.
- **MoiréLens:** Co-first-author work on gas-flow visualization using Moiré-embedded backgrounds, camera calibration, and computational readout. *ACM/IEEE SenSys 2026; MobiCom 2026* demo accepted.
- **Woven-surface deformation sensing:** First-author demo using naturally occurring Moiré patterns in everyday woven surfaces. *ACM MobiCom 2026*, demo accepted.

Haptic Interaction in Automated Driving

China Beacons Institute; Prof. Xu Sun

May–Sep. 2024

Built CARLA simulations and contact/contactless haptic prototypes to study feedback location and driver takeover performance. *ACM CHI 2025 Extended Abstracts*.

Wearable Localization and Sensor Fusion

University of Nottingham Ningbo China; Prof. Boon Giin Lee

Mar. 2022–Mar. 2024

Contributed formal analysis to IOAM and co-authored dual-foot zero-velocity detection research for inertial localization. *Remote Sensing 2022; IEEE SENSORS 2023*.

Infrared Thermal-Image Super-Resolution

China Beacons Institute; Prof. Boon Giin Lee

Oct. 2022–Jan. 2024

Developed FEGAN, a feature-oriented GAN for thermal-image texture and edge recovery. First-author article in *IEEE Signal Processing Letters 2024*.

ENGINEERING EXPERIENCE

Loctek Ergonomic Technology — Embedded Engineer

Jun.–Sep. 2022

Evaluated millimeter-wave sensing modules and integrated PCB electronics and wireless connectivity for a smart office-desk prototype.

Jinlin Technology — Product Manager & Embedded Engineer

Feb.–Jun. 2022

Built ESP32 and PCB prototypes for an interactive respiratory-sensing sleep product.

EMOLAX, XBOT PARK / Ningbo TechX — Co-founder & Project Leader

May–Sep. 2021

Prototyped sensor-responsive relaxation interfaces using Raspberry Pi and ESP32.

StoneBOX — Product Manager & UX Designer

Oct. 2019–Jun. 2021

Designed ordering and dispatch interactions for an on-campus delivery platform.

PUBLICATIONS

* Equal contribution.

Conference Papers

- [C1] **Zhu, L.**, Xing, L., Wang, R., and Sun, K. “MoiréSkin: Ultra-Sensitive Sensor–Actuator Visuo-Tactile Skin Using Moiré Patterns.” *ACM MobiCom*, 2026. **Accepted full paper.** [DOI]
- [C2] **Zhu, L.***, Wang, R.*, Rong, Y., and Sun, K. “MoiréLens: Bringing Schlieren Imaging into Real-World Environments Using Moiré Patterns.” *ACM/IEEE SenSys*, 2026. [DOI]
- [C3] Wang, C., Tan, C., **Zhu, L.**, Yang, R., and Hong, J. “Neural Active Sensing Vision and Manipulation for Cooperative Agents.” *IEEE ICBASE*, 2024. [DOI]
- [C4] Wu, R., Lee, B. G., Pike, M., **Zhu, L.**, Chai, X., and Wang, Y. “Enhancing DF-INS for Accurate Zero-Velocity Detection in ILBS: A Dual Foot Synergistic Method.” *IEEE SENSORS*, 2023. [DOI]

Journal Articles

- [J1] Ma, Y., Wu, R., Zhang, Y., Shang, Y., and **Zhu, L.** “Throughout Maximization for IRS-Assisted WPCN With Hybrid TDMA-NOMA Scheme.” *IEEE Access*, 13, 23384–23398, 2025. [DOI]
- [J2] **Zhu, L.**, Wu, R., Lee, B. G., Nkenyereye, L., Chung, W. Y., and Xu, G. “FEGAN: A Feature-Oriented Enhanced GAN for Enhancing Thermal Image Super-Resolution.” *IEEE Signal Processing Letters*, 31, 541–545, 2024. [DOI]
- [J3] Wu, R., Lee, B. G., Pike, M., **Zhu, L.**, Chai, X., Huang, L., and Wu, X. “IOAM: A Novel Sensor Fusion-Based Wearable for Localization and Mapping.” *Remote Sensing*, 14(23), 6081, 2022. [DOI]

Demonstration Papers and Extended Abstracts

- [D1] **Zhu, L.***, Wang, R.*, Rong, Y., and Sun, K. “Demo: MoiréLens: Invisible Flow Visualization using Moiré Patterns.” *ACM MobiCom*, 2026. **Accepted demo.** [DOI]
- [D2] **Zhu, L.**, Xing, L., Wang, R., and Sun, K. “Demo: MoiréSkin: Ultra-Sensitive Sensor–Actuator Visuo-Tactile Skin Using Moiré Patterns.” *ACM MobiCom*, 2026. **Accepted demo.** [DOI]
- [D3] **Zhu, L.**, Li, Z., Jin, W., Park, H., Sample, A., and Sun, K. “Demo: Towards Fine-Grained Deformation Sensing through Naturally Occurring Moiré Patterns in Everyday Woven Surfaces.” *ACM MobiCom*, 2026. **Accepted demo.** [DOI]
- [D4] Lan, R., Sun, X., **Zhu, L.**, Wang, Q., and Liu, B. “Exploring the Impact of Haptic Feedback Locations and Mid-air Haptic Technology on Driver’s Takeover Performance in Automated Vehicles.” *ACM CHI Extended Abstracts*, 2025. [DOI]

PATENTS AND PATENT APPLICATIONS

Named co-inventor on the following records:

- [P1] **System and Method for Detecting Gas Movements.** U.S. provisional application **64/056,530**, filed May 4, 2026.
- [P2] **An Infrared Image Generation Method, Apparatus, Device, and Storage Medium.** **CN116503494B**, Chinese invention patent; grant publication Feb. 10, 2026.
- [P3] **A Training Method for Infrared Thermal-Imaging Super-Resolution Networks with Enhanced Texture Information.** **CN116883244A**, published Chinese application, Oct. 13, 2023.
- [P4] **A New Type of Infrared Thermal Imaging Glasses.** **CN219590611U**, Chinese utility model; grant publication Aug. 25, 2023.

ACADEMIC SERVICE AND HONORS

Reviewer (2026): ACM IMWUT; IEEE Transactions on Systems, Man, and Cybernetics: Systems.

Honors: Wang Kuo Tong Memorial Fellowship, University of Michigan; Provincial Outstanding Graduate, Zhejiang, China.